

Sensor-Calibrated Rover Stop

Calibrate a distance sensor and program a rover to stop within a defined target zone.

Intermediate	1 class period	Practice & Review
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Engineering Context

An autonomous service rover must stop at a docking station without striking it or leaving too large a gap.

What You Will Practice

- Collect sensor calibration data
- Use conditional logic
- Evaluate accuracy and repeatability
- Tune a control threshold

What You Need

- VEX rover or approved mobile platform
- Distance sensor
- Programming environment
- Tape measure
- Target wall
- Engineering notebook

Student Instructions

- 1 Record sensor readings at known distances across the operating range.
- 2 Create a calibration table and identify any offset or unstable region.
- 3 Program the rover to approach slowly and stop at the initial threshold.
- 4 Run five trials from the same start point.
- 5 Measure final stopping distance for every trial.
- 6 Adjust speed, threshold, or confirmation logic.
- 7 Run five final trials and calculate average error and range.

Engineering Requirements

- Stop inside the instructor-defined target zone
- Complete five consecutive final trials
- Use sensor data rather than a fixed travel time
- Include a controlled low-speed approach

Constraints & Safety

- Use a clear test lane
- Install an emergency stop or immediate control method
- Do not run the rover toward people
- Keep sensor face unobstructed

What You Submit

- Calibration table
- Program with comments
- Initial and final trial data
- Average error and range
- Tuning reflection

Reflection Questions

- 1 Why did stopping distance vary between trials?
- 2 How did approach speed affect error?
- 3 What sensor readings should be ignored?
- 4 How could the rover detect a missing docking station?

Engineering Tip Use several consecutive valid readings before declaring that the stop threshold has been reached.	Common Mistake Using one sensor reading can make the rover react to noise or a brief reflection.
Stretch Goal Add a second phase that aligns the rover after stopping.	Career Connection Controls engineers calibrate sensors and tune approach logic for autonomous vehicles, docking systems, and robotic manipulators.